

Explaining DiaCausal — from start to end

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Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

This handbook is for **the guide meeting (Tuesday 29 September)** and **Progress Presentation-II (Wednesday 30 September)**. It explains everything we have built, in the order we built it. For each part it gives what it does, how it works, why we chose it, where the code is, how we know it works, and the question you are most likely to be asked.

Rules while presenting

1. Say **"synthetic data"** every time you show a result. No real patient data has been used.
2. Never invent a number. Every number here comes from a file in `results/` or `docs/TESTING.md` (see the [numbers card](#)).
3. Say what works **today**, what comes **next**, and **where the code is**. "Not built yet" is a good answer: it shows you know exactly where the edges are.
4. Safety first: the doctor decides, safety rules run before any estimate, doses are never shown, and the system says "insufficient evidence" instead of guessing.

1. Pitches

One line.

DiaCausal helps a doctor choose the second diabetes drug after metformin by estimating, for this one patient, what each option would do. Safety rules run first, every number has a 95% range, and every explanation cites an approved source.

30 seconds.

After metformin, Indian doctors usually add one of three drug classes: an SGLT2 inhibitor, a DPP-4 inhibitor or a sulfonylurea. Old records mislead, because different kinds of patients get different drugs. DiaCausal first removes unsafe options using cited drug-label rules. It then uses causal inference to estimate each remaining drug's six-month HbA1c change for this patient, with a 95% range, plus weight change and low-sugar risk. It explains the answer with quoted, cited passages from WHO and FDA documents. When the evidence is too thin it says so instead of guessing. It runs on a phone, behind approved sign-in with an authenticator app, and the doctor always decides.

2 minutes (for the guide).

1. **Problem.** Choosing the add-on to metformin is a *what-if* question: what would happen to this patient under each drug? Ordinary machine learning answers "what happened to

people like this", which is confounded by who got which drug.

2. **Method.** On a synthetic, India-calibrated cohort where we know the true answers, we use:
 - propensity scores, to check that similar patients got each drug (overlap);
 - doubly robust estimators (AIPW for averages, the DR-learner per patient), each with a 95% interval. Every generator number has a source and a status.
3. **Safety.** Ten cited rules (nine from FDA drug labels, one team rule) run before any estimate. An excluded drug is never estimated. Doses are never shown. Propensity below 0.05, or a 95% range wider than 1.5 points, means "insufficient evidence".
4. **Evidence (RAG).** The system searches only licence-cleared documents: WHO 2018 and seven FDA safety communications. It quotes the sentences that answer the question, with citations, and refuses questions the sources don't cover and any dose question.
5. **Proof.**
 - The benchmark runs 20 cohorts × 5,000 patients: the naive comparison is off by −0.159 HbA1c points, AIPW by +0.002.
 - All 9 refutation tests pass.
 - The RAG test set: recall@5 0.933, citation precision 1.0, zero dose leaks.
 - Tests: 202 for the engine, RAG and website, plus 647 for the chat app. They run on Linux, macOS and Windows.
6. **Where it runs.**
 - A phone website (diacausal.netlify.app) with admin-approved accounts.
 - A Streamlit demo and an API.
 - A secure chat application with CAPTCHA and MFA sign-in, guards, a patient panel, clinical guardrails, voice input and Docker deployment. Since 29 September its pipeline runs the causal engine and the evidence search too (all six stages).
7. **Next (October).**
 - A better search model (medical embeddings + reranker), target 3 October.
 - A real-data adapter and evaluation tools, target 6 October (100% implementation).
 - Doctor review of the test questions and the clinician vignettes.
 - Results frozen on 16 October; then the paper.

An everyday analogy (for non-technical listeners).

Think of a careful senior doctor with a pharmacist beside them. The pharmacist first crosses out any drug the patient must not take, reading the reason from the label. The senior doctor then compares the remaining drugs only against patients who really were similar. They give a best guess *with its uncertainty*, or say "we haven't seen enough patients like this one". A librarian hands over the exact pages that support each point. The junior doctor, our user, makes the decision.

2. The story from start to end

When	What we built	Where / proof
July – August	Topic, proposal, literature. First research code: a 2-drug causal notebook, a small guideline retrieval (<code>causal_engine/</code> , <code>notebooks/</code> , <code>RAG/</code>). An early RAG prototype on the open-source Kotaemon framework (repo <code>DiaCausal-RAG-Core</code>).	Initial commit 27 Aug

When	What we built	Where / proof
Early September	Data decision. The Pima dataset is not Indian and not about drug choice, so we rejected it. We chose a synthetic, India-calibrated cohort where the true effect of every drug is known. Scope frozen: adults with type 2 diabetes on metformin; three add-on classes; outcome = 6-month HbA1c change.	docs/01_Master_Plan.md
18 – 19 Sept	Chat app: walking skeleton. React page → FastAPI server → six-stage pipeline → reply, with one trace ID on every log line. One command checks everything.	PR #1, #2; docs/explain/01
20 Sept	Guards. Foul language, patient identifiers, out of scope, emergency, language. The same rules file is used in the browser and on the server.	shared/guard_rules/; docs/explain/03
20 Sept	Sign-in. CAPTCHA, Argon2id passwords, authenticator-app MFA, sessions, CSRF, logout, audit log, admin command line.	backend/app/auth/; docs/explain/04
20 Sept	Voice input. Speech-to-text runs on our own computer (faster-whisper) and puts the text <i>in the box</i> , never sends it.	docs/explain/12
20 – 21 Sept	Patient panel (structured facts with units and plausibility ranges) and clinical guardrails (15 cited rules, applied before any ranking).	PR #5, #6; docs/explain/05, 06
20 – 21 Sept	Deployment. One Docker image, HTTPS through Caddy, security headers, reachable from a phone. Evaluation pack: 25 synthetic vignettes plus the SUS usability form.	PR #7; docs/DEPLOY.md, eval/
25 Sept	Causal engine v0.3 , built in ten steps: a) cohort → b) safety rules → c) propensity → d) naive/IPW/matching/AIPW → e) DR-learner → f) metrics → g) benchmark → h) Causal Output + Streamlit → i) API → j) invariant tests.	PR #11; docs/explain/07
25 Sept	Refutation tests and E-values ; the first RAG skeleton with its licence table.	PR #12
25 – 26 Sept	Phone website running the same engine in the browser (parity-tested against Python). Accounts: Supabase, authenticator MFA, admin approval. Evidence tab (first FDA source).	PR #13, #14; https://diacausal.netlify.app
26 Sept	Mid-sem kit. Report (LaTeX), slides, Gantt chart, backup video, viva sheet.	PR #15; docs/midsem/
26 Sept	Engine extras: weight change and hypoglycaemia risk (each with a 95% range), "too uncertain" abstention, printable consultation summary. Full benchmark rerun; the primary numbers are unchanged.	PR #16
26 Sept	Full RAG. WHO 2018 plus seven FDA safety communications; sentence-aware chunks; dose text kept out of the index; coverage abstention; explanations in three modes (quoted / Gemini free tier / local Ollama) behind one citation checker; a 60-question test set and evaluation; the Supabase Edge Function for Gemini; evidence on every option card.	PR #16; website redeployed

What "25% implementation" means for us. The mentor asked for about 25%. Every core part now works on its own: the causal engine, the evidence search with explanations, the website with accounts, and the secure chat app. The largest remaining steps are:

- joining the engine and RAG into the chat app's pipeline;
- the doctor's review and the usability study;

- a better search model;
- the paper.

3. Every component, explained

Each part follows the same pattern: **What, How, Why, Where, Proof, Likely question**.

Part A — The chat application (the secure front door)

A1. Walking skeleton and trace IDs

- **What:** A doctor types a question. The page checks it, labels it with an 8-character trace ID and sends it to our Python server. The server runs six stages in a fixed order:

1. `backend_guard`
2. `clinical_guardrails`
3. `causal_engine`
4. `rag_retrieval`
5. `llm_explanation`
6. `output_guard`

The reply comes back with the same ID.

- **How:**
 - React + Vite + TypeScript on the page; FastAPI + Pydantic v2 on the server.
 - The API accepts exactly one JSON shape (`schema_version`, `client_trace_id`, `parts`). It refuses unknown fields, and anything wrong gets a 422 with a plain-English message.
 - Every log line carries the trace ID and **never the message text**.
- **Why:** Joining the pieces is where projects break, so we joined them first, while simple.
- **Where:** `backend/app/main.py`, `backend/app/schemas.py`, `backend/app/pipeline/`, `frontend/src/lib/chatFlow.ts`.
- **Proof:** `python3 scripts/check_all.py` prints 43 pass/fail lines, including 401 backend tests, 246 frontend tests and live server checks. GitHub runs it on Linux, Windows and macOS.
- **Q:** *"Do all six stages really run in the chat app?"* **A:** Yes, since 29 September (implementation Section 3). We built and validated the engine and RAG as standalone modules first, then plugged them into the stage slots that were already there. `causal_engine` runs only on the options the guardrails allow, and says which patient detail is missing instead of guessing; `rag_retrieval` and `llm_explanation` quote cited sentences; the output guard withholds anything that looks like a dose (`backend/tests/test_engine_stages.py`).

A2. Guards

- **What:** The guards block foul language, patient identifiers (notice 09), out-of-scope questions (10), emergencies (11) and unsupported languages (12). They never show or log the word they matched.
- **How:**

- One rules file, `shared/guard_rules/rules.v1.json`, is loaded by both the browser and the server. The server has the final say, because a request can skip our page.
- Both sides must give the same answer on 47 example cases plus 55 extra tricky cases.
- The profanity list comes from LDNOOBW (CC BY 4.0), with clinical words removed and a medical allowlist added.
- **Where:** `frontend/src/lib/guards.ts`, `backend/app/pipeline/blocklist.py`; `docs/explain/03-guards.md`.
- **Q:** *"Why check twice?"* **A:** The browser check is for speed and friendliness. The server check is for security, because anyone can call the API directly.

A3. Sign-in with CAPTCHA and MFA (the "master login")

- **What:** Accounts are created by an admin. Sign-in needs a user ID, a password, a CAPTCHA, and then a 6-digit authenticator code.
- **How:**
 - Passwords are stored only as Argon2id hashes.
 - The CAPTCHA is an image plus audio, made offline by the Python `captcha` library.
 - MFA uses TOTP (Google/Microsoft Authenticator); its secret is encrypted in the database.
 - Sessions use `__Host-` cookies (HttpOnly, Secure, SameSite=Strict) with a CSRF token.
 - Idle timeout 15 minutes; hard limit 8 hours.
 - Five failures lock the account for 15 minutes.
 - An audit log records sign-ins and every chat request (IDs and outcomes only).
 - Built to OWASP ASVS level 1 and NIST SP 800-63B.
- **Where:** `backend/app/auth/`; `docs/explain/04-login-and-mfa.md`.
- **Q:** *"What is the master login?"* **A:** An admin *role*, not a shared password. Admin actions need the admin's own password plus a current code.

A4. Patient panel

- **What:** Age, diabetes duration, HbA1c, eGFR, BMI, yes/no histories and budget are entered in separate boxes with their units. They are sent as a structured `patient` part next to the question.
- **Why:** A parser guessing values from free text ("sugar 8.4", "no DKA hx") would change the recommendation. Structured facts are deliberate and checkable. An empty box means *unknown*.
- **How:**
 - Plausibility ranges live in `app/patient_ranges.py` and `features/patient/ranges.ts`; a test compares the two.
 - BMI uses the Asian-Indian cut-offs: overweight from 23, obese from 25.
 - Patient values are **never logged**.
- **Where:** `docs/explain/05-patient-panel.md`.

A5. Clinical guardrails (chat app)

- **What:** 15 cited rules (14 in use) decide **before any ranking** which options may be used. A "do not use" option is removed, so nothing later can rank it first.
- **How:**
 - Thresholds live only in `backend/app/clinical/guardrails.v1.yaml`, each with a source.

- A rule marked TODO, or without a source, never fires.
- **Q:** "The engine has its own rules table — why two?" **A:** They were written at different times from different sources (`docs/CAUSAL_PLAN.md` §4). When they are joined, the stricter action wins until the doctor decides.

A6. Voice input

- **What:** Press the mic, speak, and the words appear **in the message box**. They are never sent automatically.
- **How:**
 - faster-whisper (the `base` model, int8) runs on our own server.
 - A prompt that names the drugs and units improves recognition.
 - A careful tidy-up step ("satagliptin" → sitagliptin, but never one real drug turned into another).
 - The audio is deleted after decoding, and the transcript is never logged.
- **Why:** A spoken question can contain patient details, so no cloud service is used.

A7. Deployment and evaluation pack

- **What:** `docker compose up --build -d` serves the page and the API from **one origin**, with HTTPS (Caddy) and strict security headers. `/docs` is off in production.
- **Why:** With one origin there is no CORS to get wrong, and the secure cookie works as designed. HTTPS is required for the microphone and for secure cookies.
- **Evaluation pack:** `eval/` holds 25 synthetic patient vignettes and the SUS usability and feedback forms, for the clinician study in October.

Part B — The causal engine (the heart of the project)

B1. The question, and why it is causal

- **What:** For one adult on metformin, what would the 6-month HbA1c change be under an SGLT2 inhibitor, a DPP-4 inhibitor or a sulfonylurea?
- **Why causal:** Old records mix the drug's effect with *who got it* (confounding by indication).
Worked example (made-up numbers):
 - Pooled, SGLT2i vs SU looks like **−0.425**.
 - Comparing like with like, it is **−0.15**, about 3× smaller.
- **Say:** "Ordinary machine learning predicts what will happen. We need what *would* happen under each drug for the same patient."

B2. Synthetic, India-calibrated cohort with a known truth

- **What:** We simulate patients already on metformin:
 - age, sex, diabetes duration, HbA1c, eGFR, BMI;
 - heart disease, heart failure, past hypoglycaemia, past DKA, past pancreatitis, low income.

Each patient gets a drug the way doctors tend to choose (confounding built in on purpose). We store the outcome under *all three* drugs, so the true answer is known.

- **Rules:**

- Every number in `data/params.yaml` has a `source` and a `status`: CITED, ASSUMED-DIRECTIONAL or TEAM-SET. The loader refuses anything else.
- A causal diagram (DAG) fixes which variables we adjust for, and never a mediator such as weight change.
- **Why synthetic:** Pima is not Indian data and has no drug choice. Real Indian records need ethics approval. A known truth is the only way to *prove* a method is right.
- **Where:** `diacausal_engine/cohort.py`, `data/params.yaml` (including its `dag` section, read by `diacausal_engine/dag.py`).
- **Q:** *"Doesn't synthetic data prove nothing?"* **A:** It proves the *method* recovers a planted truth under realistic confounding, which you cannot check on real data. It does not prove real drug effects, and we never claim it does. `cohort.load_dataset()` already reads a real de-identified CSV once approved.

B3. Safety rules first

- **What:** Ten rules (R01-R10), each with its source. Nine come from the US FDA labels of dapagliflozin (FARXIGA), sitagliptin (JANUVIA) and glimepiride; R10 is a team rule pending clinician review. Three are EXCLUDE (R01, R02, R10); seven are CAUTION.
 - R01: SGLT2i not for glycaemic control at eGFR < 45.
 - R05: DPP-4i with past pancreatitis.
 - R07-R10: sulfonylurea and hypoglycaemia risk or kidney function.
- **How:** Rules run **before** any estimate. An excluded option is never estimated. Thresholds live only in `data/rules.csv`, and a test scans the code for hard-coded thresholds.
- **Doses:** never shown. An output check refuses any reply containing dose-like text.

B4. Propensity score and overlap

- **What:** The chance that a patient like this one received each drug. It comes from a multinomial logistic regression, cross-fitted over 5 folds and clipped for stability.
- **Why:** If almost nobody like this patient received a drug (propensity below **0.05**), there is no fair comparison, so the answer is **"insufficient evidence"**, never a number.
- **Balance proof:** before weighting, the groups differ by up to |SMD| 0.704; after IPW weighting, about 0.08 (0.0775 as stored). Under 0.1 counts as balanced.

B5. Average effects: naive, IPW, matching, AIPW

- **Naive:** compare group means. Biased on purpose, to show the problem.
- **IPW:** weight each patient by 1/propensity, so each drug group looks like the whole population.
- **Matching:** pair patients with similar propensity. Intervals come from a bootstrap.
- **AIPW (doubly robust):**
 - Estimate = outcome-model guess + an IPW correction of that guess's error.
 - It is correct if **either** the outcome model **or** the propensity model is right.
 - Its 95% interval comes from the influence function.

B6. Per-patient effects: DR-learner

- **What:** The expected change for *this* patient under each drug, plus the differences between drugs, each with a 95% interval.

- **How:** Regress the AIPW scores on the patient's features. The second stage is linear, and the intervals come from HC3 robust standard errors.
- **Q:** *"Why not a bigger model?"* **A:** Every number must carry a valid interval, and must be exactly reproducible in the phone browser. A linear second stage on doubly robust scores gives both.

B7. Secondary outcomes and "too uncertain" (added 26 Sept)

- **What:** Each estimated option also shows:
 - the **6-month weight change (kg)**, with a 95% range;
 - the **risk of any hypoglycaemia (%)**, with a 95% range.

They come from the same cohort and method, generated from a separate random stream, so the HbA1c results are byte-for-byte unchanged.

- **Too uncertain:** if a 95% range is wider than `engine.max_interval_width = 1.5` points (TEAM-SET, for the doctor to confirm), the option shows "insufficient evidence (too uncertain)".
- **Proof:** see the [numbers card](#).
 - Weight, SGLT2i vs SU: truth -3.33 kg; estimate -3.33 .
 - Hypoglycaemia, SU vs DPP-4i: truth $+12.9$ points; estimate $+12.6$ ($+12.55$ in the file).

B8. Benchmark, refutation and sensitivity

- **Benchmark:** 20 fresh cohorts $\times$ 5,000 patients, plus 2,000 test patients. We measure bias, RMSE, 95% coverage, PEHE (per-patient error), policy regret and abstention.
- **Refutation (all 9 pass):**
 - a placebo drug should show about zero effect;
 - adding a random common cause should change nothing;
 - an 80% subset should give about the same answer.
- **E-values:** how strong a hidden confounder would need to be to explain the effect away: 1.72 (SGLT2i vs DPP-4i), 2.15 (SU vs DPP-4i), 1.57 (SGLT2i vs SU).
- **Where:** `python -m diacausal_engine.benchmark, results/`, five figures in `results/figures/`.

B9. Causal Output, Streamlit demo and API

- **Causal Output:** the structured JSON handed to the evidence-fusion layer (`schemas.CausalOutput`):
 - APPLICABLE / NOT_APPLICABLE;
 - intervention and outcome;
 - per-option status (estimate / excluded / insufficient evidence) with the effect, its interval, the propensity, the fired rules with sources, and the secondary outcomes;
 - comparisons, assumptions and versions;
 - the intended-use sentence.
- **Streamlit demo:** `streamlit run demo/streamlit_app.py`.
- **API:** `POST /api/v1/recommend` (FastAPI) returns the same object. Every request writes an audit line with IDs, versions and rule IDs only, never patient values.

Part C — The phone website and accounts

C1. The website (<https://diacausal.netlify.app>)

- **What:** A phone-first installable web app (PWA) with these tabs:
 - **Try it:** the patient form, 3 presets, colour-coded option cards, a 95%-range chart, fair comparisons, a print summary;
 - **Evidence:** RAG;
 - **Results:** the benchmark;
 - **Learn;**
 - **About.**
- **How:**
 - The fitted model is exported to `web/model.json`, and `web/engine.js` repeats `recommend.py` step by step *in the browser*, so patient details never leave the phone.
 - Tests compare it with Python on 155 patients, including forced "too uncertain" cases.
- **Colours:** red = excluded by a cited rule; amber = caution; grey = insufficient evidence; pine = estimate with its 95% range. Nothing is ever green for "recommended".
- **Security:** a strict Content-Security-Policy (no inline scripts); no API keys in the repository.

C2. Accounts (Supabase)

- **What:** Sign up → set up an authenticator app → wait for an admin to approve → tick the intended-use box. Only approved accounts can open Try it and Evidence.
- **How:**
 - Row-level security: each person sees only their own profile.
 - Admin functions require an authenticator-verified (aal2) session.
 - Rhian is the admin and approves teammates from "Approve accounts (admin)".
- **Honest limit:** sign-in controls who can *use* the app. The model file itself is public, which is fine: it contains no patient data.

C3. Printable consultation summary (added 26 Sept)

- "Print or save as PDF" produces one clean page:
 - the patient as entered and the date;
 - the three options with ranges, weight and low-sugar risk;
 - the rule sources;
 - the versions and the intended-use sentence.
- There are no doses. Nothing is uploaded.

Part D — RAG: evidence with citations (added 25-26 Sept)

D1. Licence gate: what we are allowed to use

- **What:** `RAG/sources.csv` lists 23 candidate sources plus 3 excluded ones, each with its licence, evidence and bucket. **Only rows in the `cleared_ingest` bucket that a team member has confirmed are read.** Everything else is refused in code and by tests.
- **In the index today (8 documents, 78 passages):**
 - **S01 WHO 2018** second- and third-line medicines guideline (CC BY-NC-SA 3.0 IGO);
 - seven **FDA Drug Safety Communications** (US government works):
 - metformin and kidney function;
 - saxagliptin/alogliptin and heart failure;

- DPP-4 inhibitors and joint pain;
- SGLT2 inhibitors: ketoacidosis and urinary infections;
- acute kidney injury;
- Fournier's gangrene;
- the canagliflozin amputation warning.
- **Never ingested:** IDF 2025, ADA, NICE, KDIGO, RSSDI 2022 and textbooks (not licence-cleared).
- **Q:** "Why not just use ADA or IDF guidelines?" **A:** Their licences don't allow copying them into a tool. We cite them, but we don't ingest them.

D2. Ingestion and chunking

- **How:**
 - PDFs are converted with page markers and headings (`diacausal_rag/pdf_text.py`). FDA pages come from the official page's archived copy (`scripts/fetch_fda_dsc.py`).
 - Text is cut into passages of about 400 words that **never cross a section and end at a sentence**.
 - Every passage carries its source, version, section and page.

D3. Search: hybrid + fusion + abstention

- **How:**
 - **BM25** (exact words such as "eGFR") and **TF-IDF vector** search (similar wording) each rank the passages.
 - **Reciprocal rank fusion** merges the two rankings; the best 5 are kept.
- **Abstain** ("insufficient evidence") when:
 - the best keyword score is weak; or
 - the passages contain **less than 60%** of the question's words (`min_query_coverage`, TEAM-SET).
- **Doses:** dose-like text is **removed from the index itself**, and any passage containing a dose is withheld.

D4. Explanations (step 5 of the RAG pipeline)

- **Three modes behind one rule, "cite only the retrieved passages, otherwise abstain":**

Mode	Where	Cost	What it does
Template (default)	everywhere, offline	free	Quotes the sentences that best answer the question, each with its passage number. It cannot invent anything.
Gemini free tier	website, approved users only	free	Rewords the passages in plain English. It receives only the question and the passage IDs; our server rebuilds the passages. Never patient details.
Ollama local model	laptop / hospital computer	free, offline	The same, with nothing leaving the machine.

- **Citation checker:** runs on every model answer. Each sentence must:
 - end with a citation to a passage that was shown;
 - share most of its words with that passage;

- contain no dose.

If any sentence fails, the quoted answer is shown instead, with a note saying why.

- **Dose questions** ("What dose of glimepiride...?") get: "DiaCausal never gives doses."
- **Where:**
 - `diacausal_rag/explain.py` and its browser mirror `web/explain.js`;
 - `supabase/functions/explain/` (the key is a Supabase secret; only approved, MFA-verified users can call it).

D5. Evidence fusion on the Try-it cards

- Each option card has "**Evidence for this option**": the licence-cleared passages about that drug class and its fired rules. For example, DPP-4 inhibitor → the FDA joint-pain communication.

D6. Evaluation on a gold question set

- **Test set:** `eval/rag_gold.csv`, 60 questions:
 - 45 answerable, each with the source and section that answers it;
 - 10 out of scope;
 - 5 dose requests;
 - 20 of them flagged for the doctor to review.

- **Results** (`results/rag_eval_summary.csv`, template mode):

Measure	Value	Team target
Recall@5 (right source and section in the top 5)	0.933	≥ 0.80 met
Answerable questions answered	0.978	—
Citation precision	1.000	≥ 0.95 met
Dose leaks	0	0 met
Out-of-scope questions refused	0.800	≥ 0.95 not met

- **Honest notes:**
 - The thresholds were tuned on this same set, so these numbers are optimistic.
 - The two out-of-scope misses are a gestational-diabetes diet question and a statin question. They are the reason for the October work: a medical embedding model and a reranker.

Part E — Quality and safety across everything

Suite	Count	How to run
Causal engine	150 tests	<code>.venv/bin/python -m pytest tests/engine -q</code>
RAG	33 tests	<code>.venv/bin/python -m pytest tests/rag -q</code>
Website (incl. browser parity)	19 tests	<code>.venv/bin/python -m pytest tests/web -q</code>
Chat app backend / frontend	401 / 246 tests	<code>python3 scripts/check_all.py</code> → "ALL 43 CHECKS PASSED"

Some tests enforce project rules rather than features:

- the intended-use sentence on every reply;
- no doses anywhere;
- every number has an interval;
- no secrets in the repository;
- exact version pins;
- no clinical threshold typed into code.

No test was ever weakened to make it pass.

4. The guide meeting (Tuesday 29 Sept), about 15-20 minutes

Goal: show that far more than 25% works, and leave with decisions only the guide can make.

Min	Who	Do	Say
0-2	Pratham	The one-line and 30-second pitches	"We've built the engine, the evidence search and the website; this is what we'll present."
2-6	Rhian	Laptop: <code>.venv/bin/python -m pytest tests/engine tests/rag tests/web -q</code> (start it first, it takes about 4 minutes), then open <code>results/results_table.tex</code> and <code>results/figures/ate_vs_truth.png</code>	Naive vs AIPW against the known truth; the 9/9 refutation tests; E-values
6-10	Graceton	Phone or laptop: the live website → presets 1-3 → weight and low-sugar lines → "Evidence for this option" → print summary	The colour code; the rule sources; "insufficient evidence"
10-13	Advik	Evidence tab: "SGLT2i and ketoacidosis" → quoted explanation; "Not in the sources" → abstains; a dose question → refused	Licence gate; citation checker; 0 dose leaks
13-15	Rhian	<code>docs/midsem/</code> : the slides, report and Gantt chart	What changed since the report draft
15-20	All	Ask the guide (below)	—

Decisions to ask the guide for

1. **A doctor.** Needed for three things:
 - review of the 20 flagged test questions (`eval/rag_gold.csv`, column `doctor_review`);
 - the 25-vignette clinician review with the SUS form (`eval/`);
 - confirming the rule cut-offs.
2. **Thresholds.** Is 1.5 HbA1c points a sensible "too uncertain" limit? Is 60% question coverage a sensible "refuse" limit? Both are TEAM-SET, one line each (`data/params.yaml`, `diacausal_rag/config.yaml`).
3. **Sources.** Can the department help obtain written permission for the ICMR / RSSDI 2022 guidelines? Today they are cite-only.
4. **Prices.** Is using the Jan Aushadhi list acceptable? Until it's confirmed, the app shows "price unavailable".
5. **Final deliverable.** Is the demo the website, the chat app with the engine plugged in, or both? This decides where October effort goes.

6. **The paper.** Target venue and timeline (results frozen on 16 October).

5. Panel script (Wednesday 30 Sept, 12-15 minutes)

Keyed to the final deck `Intelligent_Diabetes_CDSS_MidSem_updated.pptx` (35 slides + 5 backup). The owners match the speaker notes; the full words for every slide are in `TEAM_BRIEFING.md`. Speak to the panel, not the screen; about 20-25 seconds per design slide, longer on 23 (the 25%) and the demo.

#	Slide	Who	Say (short version)
1-2	Title, team	Pratham	"Group 28; an Intelligent Diabetes Clinical Decision Support System; guide Mr. Rahul Jadhav."
3	Outline	Pratham	Read the headings; stress scope, interaction-sheet timeline and working of the system.
4	Problem	Rhian	The worked example: naive -0.425 vs fair -0.15 . "Different patients get different drugs, so a naive comparison mixes the drug with who got it."
5	How it works	Rhian	Seven steps, all running in the chat app since 29 Sep: patient $\rightarrow$ safety rules $\rightarrow$ overlap $\rightarrow$ estimates with 95% ranges $\rightarrow$ cost $\rightarrow$ cited evidence $\rightarrow$ the doctor decides.
6-7	Block diagram, flowchart	Rhian	The same order as boxes and as a decision flow; "insufficient evidence" is a real branch.
8	Safety rules	Advik	Ten cited rules (R01-R10) run before any number; no doses; the intended-use sentence.
9	Scope	Advik	In scope / out of scope / future scope (the guide asked for this in week 4).
10-11	Hardware & software, requirements	Advik	A laptop or phone; Python 3.12, FastAPI, React, Supabase; functional and non-functional requirements, all built.
12-13	Gantt, timeline	Advik	Only the sheet's rows. 25% signed 23 Sep; 40% met 28 Sep; 60% met 29 Sep; 80% target 3 Oct; all four sections by 6 Oct.
14-16	Architecture, data flow, use case / sequence	Graceton	Layers and flows; patient values are never logged; dashed arrows on 16 are UML return messages.
17	Working of the system	Graceton	Demo patient 2 step by step: SGLT2i excluded (R01, eGFR 40), DPP-4i -0.68 , SU -0.82 , ranges overlap.
18	Data and API	Graceton	<code>rules.csv</code> columns; <code>POST /api/v1/recommend</code> ; the Causal Output fields.
19-21	Datasets	Graceton	Synthetic cohort (why and how), NMB-2017 real Indian survey as a realism check.
22	What's built	Rhian	The status table; 202 engine/RAG/website tests + 647 chat app tests.
23	Exactly what the 25% covers	Rhian	Six steps; "at 25%, one patient goes in, unsafe drugs come out, each remaining drug gets a 6-month HbA1c change with a 95% range or 'insufficient evidence'."
24	How the code works	Rhian	AIPW = the outcome model's guess + a weighted correction; DR-learner + HC3 for one patient's range.
25-26	Website, chat app screens	Advik	The website's four tabs; the chat app's sign-in, guards, estimates and cited evidence.

#	Slide	Who	Say (short version)
27	Accuracy	Rhian	Naive off by -0.159 and never covers the truth; IPW/matching/AIPW ≈ 0.002 ; AIPW coverage 90–95%.
28	Plots	Pratham	Overlap (0.05 line), balance (all under 0.1), patient-level recovery.
29	RAG results	Rhian	recall@5 0.933, citation precision 1.0, 0 dose leaks; abstention 0.80 vs target 0.90 (said honestly).
30	Demo	Graceton (Rhian drives)	See the demo runbook .
31	Completed vs planned	Advik	Completed list; plan keyed to the sheet's weeks 8–10, guide evaluation, Nov, Dec, Jan–Feb.
32	Progress $\approx 75\%$	Pratham	The weighted table; $0.2 \times 95 + 0.25 \times 90 + 0.2 \times 75 + 0.15 \times 85 + 0.1 \times 20 + 0.1 \times 40 \approx 75$.
33	Conclusion	Pratham	Engine + cited evidence in one chat app; safety first; honest uncertainty; the doctor decides.
34	References	Advik	IEEE style; WHO 2018, FDA labels and communications, AIPW / DR-learner papers, NMB-2017.
35	Thank you	All	"Thank you. We're happy to take questions."
36–40	Backup	whoever is asked	Why synthetic is valid · real Indian datasets · path to a hospital · phone screens · literature.

Hand-overs (one sentence each, so the talk flows):

- Pratham → Rhian: "Rhian will explain why this is a causal problem."
- Rhian → Advik: "Advik will show the safety rules that run before any number."
- Advik → Graceton: "Graceton will walk through the design and our data."
- Graceton → Rhian: "Rhian will show what is built, and exactly what our 25% covers."
- Rhian → Advik: "Advik will show the screens."
- Advik → Rhian: "Rhian will show the accuracy results."
- Rhian → Pratham: "Pratham will show the plots."
- Pratham → Rhian: "Rhian will show how well the evidence search works."
- Rhian → Graceton: "Graceton will demo it live."
- Graceton → Advik: "Advik will compare completed and planned work."
- Advik → Pratham: "Pratham will show our progress and conclude."

6. Demo runbook

Before the panel (15 minutes ahead)

1. The laptop is charged, on the college Wi-Fi, with the browser zoomed to 125–150%.
2. Sign in at <https://diacausal.netlify.app>: password, then the authenticator code. The site signs out after **15 minutes idle**, so sign in close to your slot. Keep the authenticator phone with the person driving.
3. **Backup A (no internet, no sign-in)**: a local copy.

```
cd DiaCausal-CDSS/web && python3 -m http.server 8080
```

Then open <http://localhost:8080>. It shows "Local copy: sign-in is switched off".

4. **Backup B:** docs/midsem/DiaCausal_demo.mp4 (112 s, captioned), opened and paused on the first frame.
5. Only if the Gemini key has been set **and tested that morning**, the Gemini button may be shown. Otherwise don't press it; just say what it does.

Live (about 3 minutes)

Step	Click	Say
1	Try it → 1 • Typical patient	"Three estimates, each with a 95% range. SGLT2 inhibitor −0.93 (−0.98 to −0.87). Below each: weight change and the risk of any low sugar, e.g. sulfonylurea 9.9% vs DPP-4 inhibitor 1.8%."
2	2 • eGFR 40, past pancreatitis	"The SGLT2 inhibitor is red: excluded by rule R01 with its FDA label source, and never estimated. The DPP-4 inhibitor is amber: caution for R04 and R05."
3	3 • Older, past hypoglycaemia	"The sulfonylurea is grey: insufficient evidence. Almost no similar patient got it (propensity 0.006, below 0.05), so we refuse to guess."
4	Open Evidence for this option on the DPP-4 card	"Evidence fusion: licence-cleared passages about this drug class, with the source — here the FDA joint-pain communication."
5	Print or save as PDF	"A one-page consultation summary: patient as entered, ranges, rule sources, versions, no doses."
6	Evidence → SGLT2i and ketoacidosis	"The explanation quotes the FDA passages, each sentence numbered to its source."
7	Evidence → Not in the sources	"Glimepiride price in India isn't in our sources, so it says insufficient evidence instead of making something up."
8	Type "What dose of glimepiride should I start with?"	"Dose questions are always refused: doses come only from the label and the doctor."
9	Results tab	"The benchmark on synthetic data: naive vs corrected, against the known truth."

If anything fails: "Here is the same demo, recorded yesterday." Play the video. Don't apologise at length.

7. Q&A bank

The detailed answers are in docs/06_Viva_Prep.md. These are the ones most likely on the day.

Causal inference and statistics

1. **What is causal inference, in one line?** Estimating what *would* happen under each choice for the same patient, not just what happened to people who made that choice.
2. **What is confounding by indication?** Doctors choose drugs based on the patient, so the patient's condition affects both the drug and the outcome. The worked example: naive −0.425 vs fair −0.15.
3. **What is a propensity score?** The probability that a patient like this received each drug, given their details. We use it to check overlap and to weight patients.
4. **Why 0.05?** Below that, fewer than 1 in 20 similar patients got the drug. Weights would exceed 20, and the estimate would rest on a handful of people. It is the standard positivity cut-off in params.yaml, marked TEAM-SET.

5. **What does "doubly robust" mean?** AIPW combines an outcome model and a propensity model. It stays unbiased if *either one* is right.
6. **AIPW vs DR-learner?** AIPW gives the *average* effect in the population. The DR-learner gives the effect *for this patient*, by regressing the AIPW scores on patient features.
7. **How do you get a 95% interval?** For AIPW, the influence function ($\text{mean} \pm 1.96 \times \text{SE}$). For the DR-learner, HC3 robust standard errors on the second-stage regression. For matching, a bootstrap.
8. **What is coverage and why does it matter?** The share of repeats where the 95% interval contains the truth. It should be about 95%. AIPW gets 0.95; naive gets 0.00.
9. **What is PEHE?** The average error in the per-patient effect (RMSE across patients). The DR-learner scores 0.105–0.117, against 0.184–0.192 for the T-learner.
10. **The S-learner's PEHE is similar. Why not use it?** Its point accuracy is similar in this simulation, but it gives no valid 95% interval, and our rule is that every number carries one.
11. **What are refutation tests?** Checks that should fail if the method were fooling itself: a placebo drug, a random common cause, a data subset. All 9 pass.
12. **What is an E-value?** How strong an unmeasured confounder would need to be, in risk-ratio terms, to explain the effect away. Ours are 1.57–2.15.
13. **What about unmeasured confounding in real data?** That is exactly what E-values quantify. In real data we'd also add a negative-control outcome. On synthetic data, we know there is none by construction.
14. **What is policy regret?** How much HbA1c benefit you'd lose by following the model's pick instead of the truly best allowed drug. DR-learner 0.011 points vs naive 0.062.

Data

15. **Why not Pima or a Kaggle dataset?** Pima is data from Pima (Akimel O'odham) women in Arizona, USA, with no drug choice and no follow-up. It cannot answer our question.
16. **Where do the synthetic numbers come from?** Every one is in `data/params.yaml`, with a source and a status. Examples: ICMR-INDIAB, Misra 2009 (BMI cut-offs), Palmer 2016 and Tsapas 2020 (drug-effect directions), the TriMaster trial, and a team worksheet of Indian survey figures. Many are TEAM-SET, which is said openly. The loader refuses an unsourced number.
17. **When will you use real data?** After ethics approval, as a de-identified CSV. `cohort.load_dataset()` already reads it; we would rerun the benchmark checks.
18. **Is the cohort "India-calibrated"?** Its age, HbA1c, eGFR and BMI distributions and prevalences follow Indian sources, and it uses Asian-Indian BMI cut-offs (23 and 25).

Safety, ethics and regulation

19. **Is this a medical device?** No. It is a research prototype for clinician evaluation, not for unsupervised clinical use. That sentence is on every screen and every API reply.
20. **What stops a dangerous suggestion?** Cited rules run first and remove unsafe options. Doses are never shown. Low overlap or a too-wide range means "insufficient evidence", and nothing is ever green for "recommended". The doctor decides.
21. **Where do the thresholds come from?** Nine rules from FDA labels (with section numbers) and one team rule pending review; any team-set cut-off is marked so and waits for a doctor to confirm it. They are in one table (`data/rules.csv`), never in code.
22. **What about patient privacy?**
 - The website computes on the phone, and patient values are never sent.

- Logs hold IDs and outcomes, never values or message text.
 - Voice audio stays on our server and is deleted after decoding.
23. **Why no doses?** Dosing depends on details we don't model, and a wrong dose is directly harmful. Doses belong to the label and the prescriber.

RAG and LLMs

24. **What is RAG?** Retrieval-augmented generation: first *find* the relevant passages in approved documents, then answer *only* from them, citing each one.
25. **Why not just ask ChatGPT or Gemini?** They can state things that aren't in any source and can't show where each claim came from. We need every sentence traceable to a licensed page, and a refusal when there is nothing.
26. **How do you stop the model inventing things?** The citation checker. Every sentence must cite a shown passage and share most of its words with it, and contain no dose. If not, we show the quotes instead. The default mode doesn't generate at all; it quotes.
27. **BM25 vs vector search?** BM25 matches exact terms ("eGFR", "pancreatitis"). Vector search matches similar wording. Rank fusion keeps passages that either method ranks highly.
28. **Why TF-IDF and not embeddings?** It is fully reproducible in the browser today. A medical embedding model and a reranker are the October upgrade, and there is a slot for them in the code.
29. **Abstention is 0.80, below the 0.95 target. Why?** Two out-of-scope questions (a gestational-diabetes diet question, statins) share enough words with our documents to slip through. We report it openly. The fix is semantic search and a reranker, then a held-out test set.
30. **Weren't the thresholds tuned on the test set?** Yes, and we say so: the numbers are optimistic. The doctor's review and a separate held-out set are the honest next test.
31. **Licences?** Each source has a licence row. Only confirmed, cleared sources are ingested: WHO under CC BY-NC-SA 3.0 IGO with attribution, and FDA texts as US government works. ADA, IDF and NICE are cite-only.
32. **Does Gemini see patient data?** No. Only the question and the IDs of the passages go to our server, which rebuilds the passages. Only approved, MFA-verified users can call it. The key is a server secret.
33. **Offline?** Yes. The quoted explanations need no internet, and a local model (Ollama) can reword them on a laptop or hospital PC (`docs/OFFLINE_INSTALL.md`).

Software, security and engineering

34. **Why the same engine in Python and JavaScript?** So the phone computes privately. Tests compare the two on 155 patients and 100 questions, and they must agree.
35. **How are passwords and sessions protected?**
- Chat app: Argon2id hashes, TOTP MFA, `__Host-` cookies, CSRF tokens, logout, audit log.
 - Website: Supabase auth with MFA, row-level security and admin approval.
36. **How do you know it works on other computers?** GitHub CI runs everything on Linux, macOS and Windows, with exact version pins.
37. **What is the trace ID for?** To follow one message through the browser console and every server log line, without ever logging its content.
38. **Why Docker with one origin?** No CORS configuration to get wrong, secure cookies work, and HTTPS enables the microphone.

Scope and future

39. **Why only three drug classes?** They are the common oral add-ons after metformin in India. A narrow scope lets us do it properly; GLP-1 and insulin are future work.
40. **Why HbA1c at 6 months?** It is the standard glycaemic outcome in trials and guidelines. Weight and hypoglycaemia are now secondary outcomes.
41. **What is left?**
- The engine and RAG inside the chat app;
 - doctor review and SUS;
 - an embedding model and reranker;
 - results frozen 16 October;
 - the paper;
 - the final report.
42. **What would a hospital deployment look like?** An offline install on the hospital's own machines (docs/OFFLINE_INSTALL.md), a local de-identified dataset after ethics approval, and the same rules and tests.
43. **What is novel here?**
- Causal per-patient estimates *with* intervals and abstention;
 - safety rules strictly before estimation;
 - licence-gated, citation-checked evidence;
 - all of it reproducible on a phone and validated against a known truth.
44. **What did each member do?** Each person answers for their own part in one sentence. Write yours here before the day, matching the slides you present: _____
45. **What if the doctor disagrees with the estimate?** Then the doctor is right to decide. The tool shows evidence and uncertainty and records the request; it never overrides.

8. Honest limitations and the October plan

Limitations to state if asked

- **Synthetic data:** the results prove the methods recover a planted truth, not real-world drug effects.
- **Search:** it is lexical (BM25 + TF-IDF). Abstention is 0.80, below target, and was tuned on the same test set.
- **Clinical review:** no doctor has reviewed the test set or the vignettes yet.
- **Chat app:** all six stages run since 29 September, but the explanation uses the quoted template by default; Gemini / Ollama are optional and always pass the citation checker.
- **Prices:** "price unavailable" until the Jan Aushadhi list is confirmed.
- **Rule cut-offs:** some are TEAM-SET and await clinician confirmation.

October (as in the interaction sheet and the Gantt chart)

- Done early: the engine and RAG inside the chat app pipeline (29 Sep, the sheet's 60% check).
- A medical embedding model and a reranker; a held-out test set (target 3 Oct, the 80% check).
- A real-data adapter and evaluation tools (target 6 Oct, 100% implementation).
- Doctor review of the 20 flagged test questions and the clinician vignettes; the SUS usability study.

- Guide evaluation 26–31 Oct; results frozen on 16 October; the IEEE-style paper; the final report.

9. Numbers card

All results are on **synthetic data**. Each row names its source file.

Benchmark: 20 cohorts × 5,000 patients, 2,000 test patients (`results/run_info.json`, `results/benchmark_summary.csv`)

Comparison	Truth	Naive: bias / coverage	IPW	Matching	AIPW: bias / coverage
SGLT2i vs DPP-4i	−0.211	−0.159 / 0.00	+0.001 / 1.00	−0.002 / 0.95	+0.002 / 0.95
SU vs DPP-4i	−0.256	−0.106 / 0.00	−0.004 / 1.00	−0.008 / 1.00	−0.003 / 0.95
SGLT2i vs SU	+0.045	−0.052 / 0.50	+0.005 / 0.95	+0.006 / 0.95	+0.005 / 0.90

Per-patient effects and policy (`benchmark_summary.csv`)

Learner	PEHE (SGLT2i-DPP4i / SU-DPP4i / SGLT2i-SU)	Policy regret
DR-learner	0.105 / 0.117 / 0.108; 95% coverage 0.94 / 0.95 / 0.93	0.011
S-learner	0.103 / 0.104 / 0.110 (no intervals)	0.012
T-learner	0.189 / 0.192 / 0.184	0.029
Naive	0.214 / 0.197 / 0.169	0.062

Other checks

- **Balance:** largest |SMD| 0.70 before → 0.08 after IPW (0.7038 → 0.0775 in the file). Abstention in the benchmark: 2.5% of patient-option pairs.
- **Refutation** (`results/refutation.csv`): 9 of 9 pass.
- **E-values** (`results/evalues.csv`): 1.72 / 2.15 / 1.57 (confidence-interval limits 1.54 / 1.95 / 1.39).

Secondary outcomes (`benchmark_summary.csv`, sections `secondary_weight`, `secondary_hypo`)

- Weight, SGLT2i vs SU: truth −3.33 kg, AIPW −3.33 (coverage 1.00).
- Weight, SU vs DPP-4i: truth +1.50, AIPW +1.48.
- Hypoglycaemia, SU vs DPP-4i: truth +12.85 points, AIPW +12.55.
- Per-patient 95% coverage: weight 0.94–0.96; hypoglycaemia 0.93–0.94.

Demo presets (website engine, reference cohort n = 5,000)

Preset	SGLT2i	DPP-4i	Sulfonylurea
1 · Typical: 52 y, HbA1c 8.4, eGFR 88, BMI 27	−0.93 (−0.98 to −0.87); −2.1 kg; low sugar 2.5%	−0.70 (−0.76 to −0.64); −0.4 kg; 1.8%	−0.94 (−1.00 to −0.88); +1.0 kg; 9.9%

Preset	SGLT2i	DPP-4i	Sulfonylurea
2 · eGFR 40, past pancreatitis	Excluded (R01)	−0.68 (−1.09 to −0.27), caution R04, R05	−0.82 (−1.21 to −0.42), caution R09
3 · 80 y, eGFR 38, past hypoglycaemia	Excluded (R01)	−0.63 (−0.86 to −0.41), caution R04	Insufficient evidence (propensity 0.006), caution R07–R09

RAG (results/rag_eval_summary.csv ; 60 test questions; 78 passages from 8 documents)

Recall@5	Answered	Citation precision	Dose leaks	Out-of-scope refused
0.933	0.978	1.000	0	0.800

Tests (docs/TESTING.md)

- Engine 150 · RAG 33 · website 19 (= 202).
- Chat app: backend 401 · frontend 246 · `check_all` 43 checks.
- CI on Linux, macOS and Windows.

10. Glossary

Term	Plain meaning
HbA1c	Average blood sugar over about 3 months, in %. Our outcome is its change in percentage points.
eGFR	Kidney function, mL/min/1.73 m ² . Lower is worse; several rules depend on it.
SGLT2i / DPP-4i / SU	The three add-on drug classes, e.g. dapagliflozin / sitagliptin / glimepiride.
Confounding by indication	Sicker patients get certain drugs, so naive comparisons mislead.
Propensity score	The chance that a patient like this got each drug.
Overlap / positivity	Enough similar patients got each drug to compare fairly (propensity ≥ 0.05).
IPW	Inverse probability weighting: re-weight patients so the groups are comparable.
AIPW	Doubly robust average effect: outcome model + weighted correction.
DR-learner	Doubly robust per-patient effect with a 95% interval.
SMD	Standardised mean difference; below 0.1 = balanced.
Coverage	How often a 95% interval contains the truth (target about 0.95).
PEHE	Per-patient effect error (lower is better).
E-value	Strength a hidden confounder would need to explain the effect away.
DAG	A causal diagram deciding which variables to adjust for.
RAG	Find approved passages first, answer only from them, cite each.
BM25 / TF-IDF	Keyword ranking / word-weight vector similarity.
RRF	Reciprocal rank fusion: merges several rankings.
Recall@5	The right passage is among the top 5.
Citation precision	The share of explanation sentences truly supported by the passage they cite.
Abstention	Saying "insufficient evidence" instead of answering.

Term	Plain meaning
TOTP / MFA	The 6-digit authenticator code; the second sign-in factor.
RLS	Row-level security: the database only returns rows the user may see.
PWA	A website that can be installed on a phone and work offline.
TEAM-SET	A value our team chose (with its reason), to be confirmed by a clinician.

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use. All results on synthetic data.